

M36 — Nursing Station QR + instructor launch button

Phase: Phase 7 follow-on (post-M35, operator-feedback fix) **Status:** **DONE** **Blocked by:** M5 (Multi-Patient Control dashboard), M22 (Per-Patient Console), M27 (Nursing Station student role), M31 (per-encounter QR card) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Operator feedback after M35:

"On the control room or encounter control, each should have a QR code setup to launch the nursing station in a new computer or tablet, and a button to open the nurses station from the control page by opening a new window to host the nursing station for the instructor."

Two launch points for the Nursing Station from instructor surfaces:

1. **QR code** — for a separate tablet/laptop to scan and host the Nursing Station view on a different device. Encodes the room- level URL `/portal/students/join?code={room_code}` (where the scanner picks the Nursing Station role on the join form). 2. **Open Nursing Station button** — opens the Nursing Station in a new tab on the instructor's current machine. Auto-creates (or reuses) a nurse-station student named `"Instructor (Nursing Station)"` behind the scenes — same pattern as M34's EHR launch.

Both launch points appear on:

- **Multi-Patient Control dashboard** (`/portal/room`) — as a dedicated ■ Nursing Station panel between the top bar and the encounter grid.
- **Per-Patient Console** (`/portal/room/encounter/{id}`) — as a 4th cell in the QR-codes card, alongside Chat / EHR / Device station cells.

The Nursing Station is room-scoped (supervises every bed), so its QR encodes the **room code**, not any encounter's join code. The QR card's footer copy on each console explains this so the operator doesn't confuse the two codes.

2. Structure

Files touched:

- `portal/server.py`:
- `portal_room` (route GET `/portal/room`) now fetches the active room via `control_room.get_active_room()` and passes a room dict (`{room_code, room_id, label}`) + `base_url` to the template context. Both are None-safe — the template renders the nurse-station panel only when room is truthy. - New route GET `/portal/control/launch_nurse_station` — instructor convenience launcher. Looks up an existing nurse_station student with `display_name == "Instructor (Nursing Station)"`; if found, redirects to its sid URL; otherwise creates one via `room.add_student(name, role="nurse_station")` and redirects there. Bounces to `/portal/room` (303) when no room is active. - New module-level constant `_INSTRUCTOR_NURSE_STATION_NAME = "Instructor (Nursing Station)"`.
- `portal/templates/control_room.html`: new `<section`

`class="nurse-station-launch">` between the empty-state block and the encounter grid; renders only inside `{% if room %}`.

- `portal/templates/encounter_console.html`: new 4th `<div`

`class="qr-cell qr-cell-nurse">` in the QR-codes card, plus the footer copy clarifying that the Nursing Station scans the room code (not the encounter's join code).

- `portal/static/control_room.css`: new `.nurse-station-launch`,
`.nurse-station-launch-btn`, `.nurse-station-launch-qr`, `.nurse-station-qr-img`,
`.nurse-station-qr-url` styles + mobile breakpoint.

- `portal/static/encounter_console.css`: `.qr-grid` bumped to

`repeat(4, 1fr)`, added a midpoint breakpoint (≤ 1000 px $\rightarrow$ 2 columns) so 4 cells still collapse cleanly. New `.qr-cell.qr-cell-nurse` styling (green tint to distinguish from the three encounter-scoped cells).

No schema migration. No new dataclass field. The instructor nurse-station student is just a regular Student row with `role="nurse_station"` and a specific `display_name`.

3. Uses

3.1 QR-on-separate-device flow

1. Instructor on `/portal/room` (or any Per-Patient Console) sees the ■ Nursing Station QR. 2. A separate tablet/laptop scans the QR. 3. Browser opens `{base_url}/portal/students/join?code={room_code}` — the same public landing page the M27 `register_nurse` flow uses. 4. Scanner picks the **Nursing Station** role on the join form, types a display name, submits $\rightarrow$ POST `/portal/students/register_nurse` $\rightarrow$ 303 to `/portal/students/nurse_station?sid=<student_id>`. 5. The tablet now shows the supervisor view (multi-bed telemetry, alarms, intercom).

3.2 Instructor launch flow (button)

1. Instructor clicks ■ **Open Nursing Station (new window)**. 2. Browser opens `/portal/control/launch_nurse_station` in a new tab (`target="_blank" rel="noopener"`). 3. Server resolves the active room, searches the roster for a `nurse_station` student named `"Instructor (Nursing Station)"`. - First call: creates one via `room.add_student(name, role="nurse_station")`. - Subsequent calls: reuses the existing seat — no duplicate entries in the roster. 4. 303 redirect to `/portal/students/nurse_station?sid=<sid>`. 5. Instructor lands on the supervisor view without typing anything.

3.3 Side-by-side workflow

`target="_blank"` means the Nursing Station opens in its own browser tab. The instructor can drag it to a second monitor and keep the Multi-Patient Control dashboard (or a Per-Patient Console) on the primary monitor. The ■ inline button on each Per-Patient Console gives a discovery path for instructors who arrive at the QR card and want to skip the QR step entirely.

4. Functions (exported API surface)

Symbol	Where	Purpose
GET /portal/control/launch_nurse_station	portal/server.py	303 → /portal/students/nurse_station?sid=... Creates the instructor seat on first call; reuses on subsequent calls. Bounces to /portal/room if no room is active.
_INSTRUCTOR_NURSE_STATION_NAME	portal/server.py	Sentinel display name ("Instructor (Nursing Station)") used to identify the instructor seat for reuse lookups.

5. Limitations

- **Roster cap counts the instructor seat.** Adding the instructor

nurse-station seat consumes one of the 24 student station slots (M19 cap). If a classroom already has 24 students plus a 25th instructor click on the button, the route returns 409. Acceptable — classroom capacity is the limit; the instructor seat is small next to it. We could exempt instructor seats from the cap in a future M37 if the constraint bites in practice.

- **The launcher does not honor a custom display name.** Every

instructor click ends up at the same `"Instructor (Nursing Station)"` seat. If two instructors share a session, they share the seat (last writer wins on any text the seat sends back). Out of scope today; matches the M34 EHR launcher's `"Control room (instructor)"` single-seat pattern.

- **The QR points at the public role-picker page.** The scanner

must still pick "Nursing Station" from the role list. A `?role= nurse_station` URL hint that pre-selected the role on the join form would shave one click off; left for a future M37.

- **The Per-Patient Console's 4th QR cell adds visual width.** On

narrow screens the new midpoint breakpoint at 1000 px collapses the 4-cell grid to 2-cell. Verified in tests but worth eyeing on a real 8-inch tablet.

- **The dashboard panel only appears when a room is active.** The

empty-state CTA is unchanged. If an instructor wants to preview the Nursing Station before starting a room, there's no path — acceptable, the nursing station is meaningless without encounters.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_nurse_station_launch.py::test_launch_nurse_station_creates_instructor_student_and_redirects	First call creates instructor seat + 303 to sid URL	PASS	2026-05-27
...::test_launch_nurse_station_reuses_existing_instructor_seat	Repeat calls return the same sid (no roster bloat)	PASS	2026-05-27
...::test_launch_nurse_station_no_room_redirects_to_dashboard	No active room → 303 back to /portal/room	PASS	2026-05-27

Test file	Asserts	Status	Last run
...::test_launch_nurse_station_redirect_target_actually_serves	The sid URL serves 200 (not a dead link)	PASS	2026-05-27
...::test_dashboard_renders_nurse_station_panel_when_room_active	/portal/room carries nurse-station-launch, "Open Nursing Station" copy, QR data, plain URL, launcher href, target=_blank	PASS	2026-05-27
...::test_dashboard_omits_nurse_station_panel_when_no_room	No active room → panel does not render	PASS	2026-05-27
...::test_encounter_console_renders_nurse_station_qr_cell	Per-Patient Console QR card has .qr-cell-nurse with room-coded QR + "Open here" launcher link	PASS	2026-05-27
...::test_encounter_console_help_text_clarifies_room_vs_join_code	Footer copy mentions "Nursing Station uses the room code" + the actual code	PASS	2026-05-27
Full v7 suite	239 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-codé	Initial M36 implementation: new /portal/control/launch_nurse_station route; dashboard panel + per-encounter QR cell + CSS; 8 new tests	portal/server.py, portal/templates/control_room.html, portal/templates/encounter_console.html, portal/static/control_room.css, portal/static/encounter_console.css, tests/v7/test_nurse_station_launch.py (new)

8. Open questions / known issues

- **Pre-select Nursing Station role via URL hint.** The QR could

point at /portal/students/join?code={room}&role=nurse_station and student_join.js could auto-check the nurse-station radio. Small UX win for tablet scanners. Tracked for M37.

- **Display name customization for the instructor seat.** Currently

the launcher always lands on the same `"Instructor (Nursing Station)"` seat. If a classroom has two co-instructors and they want to type comments into the supervisor intercom log separately, they'd want distinct seats. Out of scope today; the M28 intercom is currently one-way nurse → bedside so the issue is theoretical.

- **The QR is room-scoped but lives on per-encounter consoles.**

An instructor scanning four different encounter consoles each shows the same nursing-station QR. That's correct (one nurse station per room), but the redundancy could confuse someone who thinks each console scans a `*different*` nurse station. The footer copy clarifies this; if confusion persists in LAN test, consolidate to dashboard-only and remove the per-encounter cell.

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